Calculate gravitational force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate a gravitational interaction with consistent units.

SUCCESS CRITERIA - I can...

1. Identify both masses and the center separation.
2. Square the distance before dividing.
3. Check units and the scale of the result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “separation” and “mass” to explain your answer. Sentence starters are optional.

separation: distance between the specified objects or centers

mass: a measure of how strongly an object resists acceleration

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A ball is dropped from rest and falls ideally for **3 s**.
1. Use g=**10 m/s²**.
2. **Find its speed** and direction.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

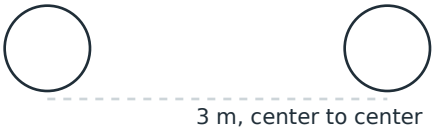
02 Worked example: explain the prediction

Reference: $F = Gm_1m_2/r^2$; $G = \mathbf{6.67\times10^{-11}\ N\cdot m^2/kg^2}$.
r is center-to-center separation. CPA may use the supplied numerator.

Worked reasoning: In the supplied gravity calculation, Gm_1m_2=**18 N·m²** and r=**3 m**. Square first: r^2=**9 m²**, so F=**18/9**=**2 N**, attractive.

Your turn: 1. **Sketch the two centers**, label r and **add the two force arrows**.
2. **Show how the units reduce** to newtons.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



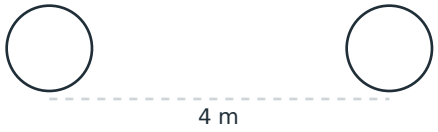
Mass 1 Mass 2

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

For spheres **4 m** apart center to center, a student substitutes r=**2 m** because each is **2 m** from the midpoint. **Name the error**.

Sentence starter: The step I would change is ... . I would change it because I found ... .



Sphere A Sphere B

Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulator supplies Gm_1m_2 = **16 N·m²**.
1. **Find the force** at each center-to-center separation in the table.
2. **Describe what happens** when the distance doubles.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Separation r (m)	r^2 (m²)	Force (N)
4	____	____
8	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why can a tiny force between classroom-sized objects be physically real even if you cannot feel it?

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **4 kg** ball is moving upward at **6 m/s** after leaving a hand. Ignore air resistance; use g=**10 N/kg**. **Draw its free-body diagram**. **Explain why each force belongs** in it.
Draw the forces on the chosen object and **label their sources**.

Word help: justify = **support your answer** with a reason or evidence

Sentence starter: I think … . I can point to …, which shows …, so … .

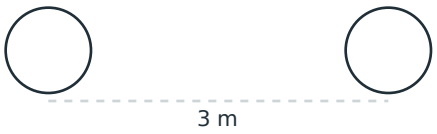
Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

Given Gm_1m_2=**36 N·m²** and r=**3 m**, find force and direction.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Keep the numerator **18 N·m²**.
2. **Find the force** at **6 m**, then design a separation that gives **18 N**.

Sentence starter: My design includes … . It does what is needed by … .

Separation (m)	Force (N)
___	___
___	___

Marzano check - circle: ① ② ③